QP CODE:25901824

MGU-BVOC (HONOURS) REGULAR EXAMINATION APRIL2026
SECOND SEMESTER
SKILL DEVELOPMENT COMPONENTS (SDC)
MG2SDCBAT101-FUNDAMENTALS OF CROP PROTECTION
(2025 Admission Onwards)

PART A

1. **Advantages of IPDM:**
   - Avoids chemical pollution of soil, water, air and food products
   - Avoids development of resistance in the plant pathogens against fungicides
   - It is an eco-friendly strategy for management of plant diseases
   - It is an economically feasible approach
   - It is a multipronged strategy for efficient management of plant diseases
2. **Gametangial copulation (gametangiogamy**): The isogametangia come in contact, their intervening wall dissolves leading to fusion of entire contents of two contacting gametangia to form a single unit.Eg. *Rhizopus stolonifer.*
3. **Weed management practices in rice fields**
   - Use of Clean and Weed-Free Seeds
   - Field sanitation
   - Crop rotation
   - Proper water management
   - Balanced fertiliser application
4. **Characteristics of a pesticide**
   - All pesticides have their own
   - Mechanism of action
   - Potency
   - Speed of effect
   - Dose required to produce effect
5. **Causes of pest outbreaks**
   - Deforestation
   - Destruction of natural enemies
   - Intensive and extensive cultivation
   - Improved agronomic practices
6. **Based on time of application**
   - Pre- plant application
   - Pre- emergence
   - Post emergence
   - Early post emergence
7. **Colonization**-Colonization starts when disease or the pathogen starts spreading in the host.
8. **Poisonous weeds**
   The poisonous weeds cause ailment on livestock resulting in death and cause great loss.These weeds are harvested along with fodder or grass and fed to cattle or while grazing the cattle consume these poisonous plants. Eg. *Datura fastuosa, D. Stramonium*
9. **Solar heat treatment (SHT)**

10. **Bacterial cell possess five shapes-** (i) spherical (Micrococcus), (ii) rod-like /bacilliform(E. Coli) and (iii) spiral-shaped (Spirillum) and (iv) curved-rod (Vibrio) and (v) club-shaped (Clavibacter).
11. **Pseudostem borer:** ***Odoiporus longicollis***

Symptoms of damage

- Grub bore into pseudostem making tunnels
- Cutting holes on outer surface
- Exudation of plant sap-initial symptom and blackened mass comes out from the bore hole
- Tunneled part decomposes and pseudostem wilts.

12. **Diseases of pulse crops:** The crops such as gram, peas, cowpea, soyabean,etc. Grown for human consumption to get protein are called pulse crops.The pathogens infect these crops and cause diseases are called the diseases-of pulse crops. Examples: Anthracnose of cowpea, wilt of gram, downy mildew of peas, leaf spot of soyabean, etc.

Diseases of cash crops: The crops such as sugarcane, cotton, jute,tobacco, etc. Grown for commercial purpose are known as cash crops. The diseases affecting these crops are called the diseases of cash crops.Examples: Red rot of sugarcane, root rot of cotton, root knot of tobacco,stem rot of jute, etc.

PART B

13. <u>Diseases of Pea</u>

<u>Fusarium wilt: *Fusarium oxysporum f.sp. pisi*</u>

Symptoms• The first symptom of the disease in the field is drooping of the plants followed by sudden death. The leaves may also turn yellow and drop off prematurely.
• Pod formation is severely affected. In collar regions of the wilted plants, necrosis and discolouration can be seen.
• The diseased plants can be pulled out from the easily than the healthy ones.
• When the diseased stem is cut, there is a dark brown,discolored band around the vascular system. Infection occurs directly through the root hairs.
• The fungus survives for indefinite periods in the soil.

Management

• Treating the seeds with Carbendazim (2 g/kg of seed) protects the seedlings during the initial stages of growth.
• Soil drenching with Copper oxychloride 0.25%.

<u>Powdery mildew: *Erysiphe polygoni*</u>

Symptoms

• The disease appears as on the foliage and pods.
• Infection is first apparent on the leaves as small slightly darkened areas, which later become white powdery spots.
• These spots enlarge and cover the entire leaf area.
• Severely infected leaves may become chlorotic and distorted before falling. Affected pods are small in size and malformed.

Management

• Spray inorganic sulphur 0.25% or Tridemorph 0.1

14. <u>Biological methods</u>

<u>Parasitism and Lysis</u>

The biocontrol against parasitizes the pathogen by coiling around the hyphae, e.g.,

Trichoderma viride; various bacteria and fungi secrete hydrolytic about the degradation of
cell wall of pathogens.e.g. (i) Bacillus sp. Causes hyphal lysis of *Gaeumanornyces graminis*.(iii) *Trichoderma* sp. Produces chitinases and β-1,3 glucanases which lyses the cell wall of*Rhizoctonia solani.*

Antibiosis

The antibiotic compounds secreted by the biocontrol agent suppress the growth of the pathogen. E.g. Phenazine-l-carboxylic acid produced by P fluorescens plays an important role in suppressing the take all disease of wheat.

Competition

The biocontrol bacteria and fungi compete for food and essential elements with the pathogen thereby displacing and suppressing the growth of pathogen.e.g. (i) the competition for nutrients between *Pythium aphanidermatum, P ultimum* and bacteria suppress the damping
off disease in cucumbers.

PGPR

Plant growth promoting rhizobacteria are bacteria that colonize plant roots, and in doing so,they promote plant growth and/or reduce disease or insect damage.

15. Weeds are plants that are unwanted in a given situation and may be harmful, dangerous or economically detrimental. Weeds are a serious threat to primary production and biodiversity.They reduce farm and forest productivity, displace native species and contribute significantlyto land and water degradation.

Characteristics of weeds

- Most of the weeds especially annuals produce enormous quantity of seeds, e.g. wild
- oats (*Avena fatua*), produces 250 seeds per plant.
- Weeds have the capacity to withstand adverse conditions in the field, because they can modify their seed production and growth according to the availability of moisture and temperature.
- They can germinate under adverse soil-moisture conditions, have short period of plant growth, generally grow faster rate and produce seed earlier than most of the crops growing in association.
- Weed seeds remain viable for longer period without losing their viability.
- Weed seeds have a tremendous capacity to disperse from one place to another through
- wind, water and animals including man.

16. **Pests of Chilli**

1.Chilli thrips: *Scirtothrips dorsalis*

Symptoms of damage:

The infested leaves develop crinkles and curl upwards
Elongated petiole
Buds become brittle and drop down
Early stage, infestation leads to stunted growth and flower production, fruit set are arrested

Management:

Inter crop with agathi (*Sesbania grandiflora*) to provide shade which regulate the thrips population
Do not grow chilli after sorghum
Grow Agathi as intercrop.
Do not follow chilli and onion mixed crop
Sprinkle water over the seedlings to check the multiplication of thrips
Treat seeds with imidacloprid 70% WS @ 12 g/kg of seed

Spray any one of the following insecticide

2.<u>Green peach aphid: *Myzus persicae*</u>

Symptoms of damage:

The infested plants turn pale with sticky appearance
The leaves curled and crinkled
Honeydew excrete – development of sooty mould
Stunted growth of the plant

Management :

Treat seeds with imidacloprid 70% WS @12 g/kg of seed
Spray any one of the following insecticide

17. <u>Red palm weevil: *Rhynchophorus ferrugineus*</u>

Symptoms of damage

Holes on trunk with with brownish ooze
Yellowing of inner leaves
Gradual wilting of central shoot in the crown

Management

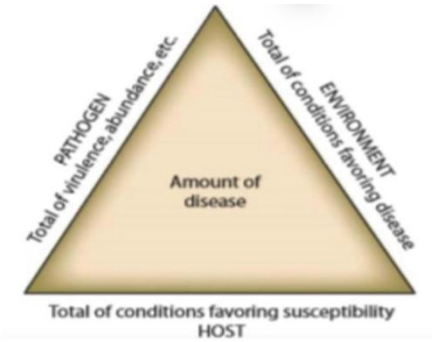
Remove and burn all wilting or damaged palms in coconut gardens to prevent further perpetuation of the pest.
Avoid injuries on stems of palms as the wounds may serve as oviposition sites for the weevil. Fill all holes in the stem with cement.
Avoid the cutting of green leaves. If needed, they should be cut about 120 cm away from the stem.
Fill the crown and the axils of top most three leaves with a mixture of Fine sand and neem seed powder or neem seed kernel powder (2:1) once in three months to preventthe attack of rhinoceros beetle damage in which the red palm weevil lays eggs.
Setting up of attractant traps

18. Disease triangle illustrating the phenomenon of plant disease as the interior space of the disease triangle is a conceptual model that shows the interactions between environment, the host and infectious agent. This model can be used to predict epidemiological outcomes in plant health, both in local and global communities.

Factors Affecting Disease Triangle

There are three factors in triangle are interacting with each other anyone from these factors is missing the triangle is not complete. The occurrence of these parameters is necessary for triangle.

If the three components of the disease triangle could be quantified, the area of the triangle would represent the amount of disease in a plant or in a plant population. If any of the three components is zero, there can be no disease. The disease triangle is also represented as a triangle with the words of the three components (host plant, pathogen, environment) placed at



the peaks of the triangle rather than along its sides.

- Host(Crop and cultivar)
- Pathogens(Bacteria, fungi, virus and nematodes)
- Environment (Temperature, humidity, pH)

PART C

19. **i).Dry Seed Treatment**

In this method, the fungicide adheres in a fine from on the surface of the seeds. A calculated quantity of fungicide is applied and mixed with seed using machinery specially designed for the purpose. The fungicides may be treated with the seeds of small lots using simple rotary seed Dresser (Seed treating drum) or of large seed lots at seed processing plants using grain treating machines. Normally in field level, dry seed treatment is carried out in dry rotary seed treating drums which ensure proper coating of the chemical on the surface of seeds.

Eg. Dry seed treatment in paddy.

Mix a required amount of fungicide with required quantity of seeds in a seed treating drum or polythene lined gunny bags, so as to provide uniform coating of the fungicide over the seeds.Treat the seeds atleast 24 hours prior to soaking for sprouting. Any one of the following chemical may be used for treatment at the rate of 2g/kg : Thiram or Captan or Carboxin orTricyclazole.

ii).Wet seed treatment

This method involves preparing fungicide suspension in water, often at field rates and then dipping the seeds or seedlings or propagative materials for a specified time. The seeds cannot be stored and the treatment has to be done before sowing. This treatment is usually applied for treating vegetatively propagative materials like cuttings, tubers, corms, setts rhizomes,bulbs etc., which are not amenable to dry or slurry treatment.

a. Seed dip / Seed soaking

For certain crops, seed soaking is essential. Seeds treated by these methods have to be properly dried after treatment. The fungicide adheres as a thin film over the seed surface which gives protection against invasion by soil-borne pathogens.Eg. Seed dip treatment in paddy.

Prepare the fungicidal solution by mixing any of the fungicides viz., carbendazim or Pyroquilon or tricyclazole at the rate of 2g/litre of water and soak the seeds in the solution for 2 hrs. Drain the solution and keep the seeds for sprouting.

b. Seedling dip / root dip

The seedlings of vegetables and fruits are normally dipped in 0.25% copper oxychloride or

0.1% carbendazin solution for 5 minutes to protect against seedling blight and rots.

c. Rhizome dip

The rhizomes of cardamom, ginger and turmeric are treated with 0.1% emisan solution for 20 minutes to eliminate rot causing pathogen present in the soil.

d. Sett dip / Sucker dip

The sets of sugarcane and tapioca are dipped in 0.1% emisan solution for 30 minutes. The suckers of pine apple may also be treated by this method to protect from soilborne diseases.

3. Slurry treatment (Seed pelleting)

In this method, chemical is applied in the form of a thin paste (active material is dissolved in small quantity of water). The required quantity of the fungicide slurry is mixed with the specified quantity of the seed so that during the process of treatment slurry gets deposited on the surface of seeds in the form of a thin paste which later dries up.

II. Soil treatment

It is well known that soil harbours a large number of plant pathogens and the primary sources of many plant pathogens happens to be in soil where dead organic matter supports active or dormant stages of pathogens.

(i) Soil drenching

This method is followed for followed for controlling damping off and root rot infections at the ground level. Requisite quantity of fungicide suspension is applied per unit area so that the fungicide reaches to a depth of atleast 10-15 cm.

Eg. Emisan, PCNB, Carbendazim, Copper fungicides, etc.

(ii) Broadcasting

It is followed in granular fungicides wherein the pellets are broadcasted near the plant.

(iii) Furrow application

It is done specifically in the control of some diseases where the direct application of the fungicides on the plant surface results in phytotoxic. It is specifically practiced in the control of powdery mildew of tobacco where the sulphur dust is applied in the furrows

.**(iv) Fumigation**

Volatile toxicants (fumigants) such as methyl bromide, chloropicrin, formaldehyde and vapam are the best chemical sterilants for soil to kill fungi and nematodes as they penetrate the soil efficiently. Fumigations are normally done in nursery areas and in glass houses. The fumigant is applied to the soil and covered by thin polythene sheets for 5-7 days and removed.

For example, Formaldehyde is applied at 400 ml/100 Sq.m. The treated soil was irrigated and used 1 or 2 weeks later. Vapam is normally sprinkled on the soil surface and covered. Volatile liquid fumigants are also injected to a depth of 15-20 cm, using sub-soil injectors.

(v) Chemigation

In this method, the fungicides are directly mixed in the irrigation water. It is normally adopted using sprinkler or drip irrigation system

20. **<u>Weeds of Coconut</u>**

Grasses

Grassy weeds dominate in well-lit coconut orchards because they propagate rapidly by stolons and rhizomes.

Imperata cylindrica **(Cogon grass)** – A vigorous perennial grass forming dense mats; difficult to eradicate due to its underground rhizomes.

Cynodon dactylon **(Bermuda grass) –** A troublesome grass that spreads through runners, competing heavily for soil nutrients.

Panicum maximum **(Guinea grass) –** Fast-growing grass that invades open plantations and

shades young seedlings.

Broadleaf Weeds

Broadleaf weeds grow mostly during the monsoon and regenerate quickly after cutting.

***Ageratum conyzoides (*Goat weed) –** A common annual herb that proliferates rapidly.

***Bidens pilosa* (Blackjack) –** Produces large numbers of seeds, causing persistent infestation.

***Chromolaena odorata* (Siam weed) –** A woody perennial shrub that aggressively colonizes
neglected coconut fields.

Sedges

Sedges thrive in moist or partially shaded areas of coconut plantations.

***Cyperus rotundus* (Purple nutsedge) –** Highly persistent due to its underground tubers; one
of the most troublesome sedge species.

1.Cultural Methods

Mulching with coconut husks, coir pith, leaves, or organic matter helps suppress weed germination and improve soil moisture.

Cover crops such as *Stylosanthes, Mucuna pruriens, Pueraria phaseoloides,* and *Calopogonium muconoides* minimize weed growth and enrich the soil.

Maintaining clean basins (1–1.8 m radius) around palms prevents weed competition.

2.Mechanical Methods

Hand weeding and hoe weeding during early stages.

Slashing and mowing of tall grasses at regular intervals.

Deep ploughing once or twice a year to destroy rhizomes and weed seeds.

3.Chemical Methods

Herbicides are used cautiously, especially in mature plantations, to avoid damage to palms.

Glyphosate (1–2%) – Effective against grasses and perennial weeds.

2,4-D (0.8%) – Controls most broadleaf weeds.

Halosulfuron-methyl – Particularly useful for sedge control

.4.Biological / Eco-friendly Methods

Controlled grazing (especially goats) to manage grassy weeds.

Use of organic mulches to suppress weed emergence naturally.

Encouraging soil microbes through compost application, which creates conditions unfavorable for weed growth.

21. **Major pest of rice**

1.Yellow stem borer: *Scirpophaga incertulas*

Damage symptoms

Larva feeds on the stem and causes drying of the central shoot known as “dead heart” in the young seedlings, and drying of the panicle in grown up plant called “white Ear”. Damage ranges from 30-80%.

Management

• Grow resistant varieties viz., Ratna, Jaya, TKM 6, IR 20 and IR 26, Sayasree, Saket, IET 3127, IET 2812, MTU 5849, PTB 12, PTB 20, PT 321, H 4

• Clip the seedling tips before transplanting to eliminate egg masses and collect & destroy the egg masses in main field.

• Avoid close planting and continuous water stagnation at early stages.

• Collect and destroy the dead hearts and white ears.

• Set up light traps to attract and kill the moths.
• Install sex pheromone traps to monitor and mass trap.
• Release the egg parasitoid, *Trichogramma japonicum* twice on 30 and 37 DAT @ 5 cc/ha/release.
• Apply *Bacillus thuringiensis* var kurstaki and neem seed kernel extract in the combination of 2.5 g/L and 1% to reduce the oviposition by the stem borer.

2.Gall midge: *Orseolia oryzae*

Damage symptoms

The maggot feeds at the base of the growing shoot causing formation of a tube like gall similar to “onion needle” or “silver-shoot”. Infested tillers produce no panicles.

Management

• Encourage early planting of the crop with quick growing varieties to escape infestation.
• Use resistant varieties like MDU-3, Shakthi, Vikram, Sureka, IR 36, Kkatiya,• Plough immediately after crop harvest.
• Remove the alternate host.
• Apply fertilizers in balanced manner.
• Set up light trap @ 1 / ha as a monitoring device. Infra red light trap attracts gall midge effectively.
• Release larval parasitoid, *Platygaster oryzae* through parasitized galls @ 1 per 10 m2 in the main field at 10 days after transplanting (DAT).

3.Swarming caterpillar: *Spodoptera mauritia*

Damage symptoms

This is a sporadic pest but causes very serious damage to young crops when it appears in large numbers. The caterpillars feed at night and hide during the day.Larvae cut the seedlings in large scale and appears as if grazed by cattle by its nocturnal feeding. Peduncles of ears are bitten through in maturing crop. They feed gregariously and march from field to field. The damage is severe in July - September. It breeds on a variety of grasses. Yield loss ranges from
10-20%.

Management

• Conserve larval parasitoids
• Conserve pupal parasitoids
• Flood the nursery to expose the hiding larvae to the surface for birds to pick them up.
• Kerosenate water during irrigation to suffocate and kill the larvae.
• Allow ducks into the field to feed on the larvae.
• Drain water from nursery and spray chlorpyriphos 20 EC 80 ml (or) endosulfan 35 EC 80 ml during late evening.

4.Rice case worm: *Nymphula depunctalis*

Damage symptoms

The caterpillars feed on green tissues of the leaves and form tubular cases around them by cutting the apical portion of leaves, which float on water. Several tubes are also seen hanging from the plants. In case of severe infestation plants are unable to grow. They damage leaf tips. The apical portion of cut leaves bear whitish papery areas since the chlorophyll is scrapped

Management

• Conserve larval parasitoids
• Conserve pupal parasitoids
• Drain water from the field
• Dislodge the cases by running a rope over the young crop
• Spray endosulfan 35 EC or monocrotophos 36 SL 1.0 L or phenthoate 50 EC 1.0 L in

500water/ha

22.Cultural methods

Eradication of alternate hosts

Removal of alternate hosts helps to prevent and check the spread of the disease caused by heteroecious rust pathogens in the primary hosts.

Eradication of collateral and self-sown overwintering hosts

There are many weed hosts or wild species of cultivated plants act as collateral hosts or volunteer plants of an economic crop which act as reservoirs of pathogens of annual crop. Reservoir hosts help the pathogen to continue the infection chain.

Crop rotation

Crop rotation is essentially a preventive measure and has its effect mainly on the succeeding crop. Crop rotation is the oldest and cheapest method adopted in agriculture for eradication of certain types of pathogens from infested soil.

Fallowing

Fallowing starves the pathogen and helps in reduction of the inoculum by elimination of the host. Diseases like Macrophomina root rot on different crop plants is controlled by following this method.

Application of organic manures

Addition of organic manures like farm yard manure or green manures or oil cakes to the soil increases the antagonistic microorganisms in the soil. Buildup of antagonistic microorganisms reduces the population of soil-borne plant pathogens and the diseases caused by them.

Soil amendment

It has been proved that the organic amendments rich in carbon and deficient in nitrogen control the take-all disease (Ophiobolus graminis) of wheat. There is considerable liberation of CO2 by soil saprophytes which suppresses the pathogenic activity of this fungus.

Selection of seeds and seed materials

Seeds and seed materials carry many fungi, bacteria, viruses and phytoplasmas and may introduce these pathogens into the field, i.e., seeds and seed materials form the primary source of infection. Seed and seed materials like cuttings, tubers, grafts, setts etc., should be well matured, disease free, uninjured and have a high germinating capacity.

Burning of stubbles and crop residues

Burning of plant wastes, crop residues, stubbles, etc., in the areas selected for raising nurseries for vegetable crops, tobacco, chillies and forest trees etc. Heats the soil and kills the inoculum of the pathogens present in the top layer of the soil

Intercropping

Intercropping is also a device in the control of some soil borne diseases. Intercrops should be properly chosen so that they should not have any common pathogen.

Barrier cropping

Taller crops can be grown to protect a crop of lesser height from virus vectors. The insects may and at the taller crops (barrier crops) and the dwarf crop may escape from virus diseases by those insects.

Roguing

Rouging consists of completely removing or uprooting the diseased plants to prevent further spread of the disease. This method is widely adopted in the control of virus diseases spread by insects

Adjustment of sowing time

In many diseases the incidence is more severe when the susceptible stage of the plant growth and favourable conditions for the pathogens coincides. While choosing the time of sowing it should be taken into consideration that susceptible stage of the crop growth and soil

conditions and other environments favourable for maximum activity of the pathogen does not fall at the same time. Properly adjusting the sowing dates can give good dividends.

Seed rate

Use of higher seed rate in the nursery creates favourable microclimate for the pathogens causing damping-off in vegetables, tobacco, chillies and forest nurseries. Hence, use of optimum seed rate should be adhered in such crops.

Spacing

Closer spacing invariably alters the microclimate underneath the canopy of the crop which may provide favourable environment for development of diseases.